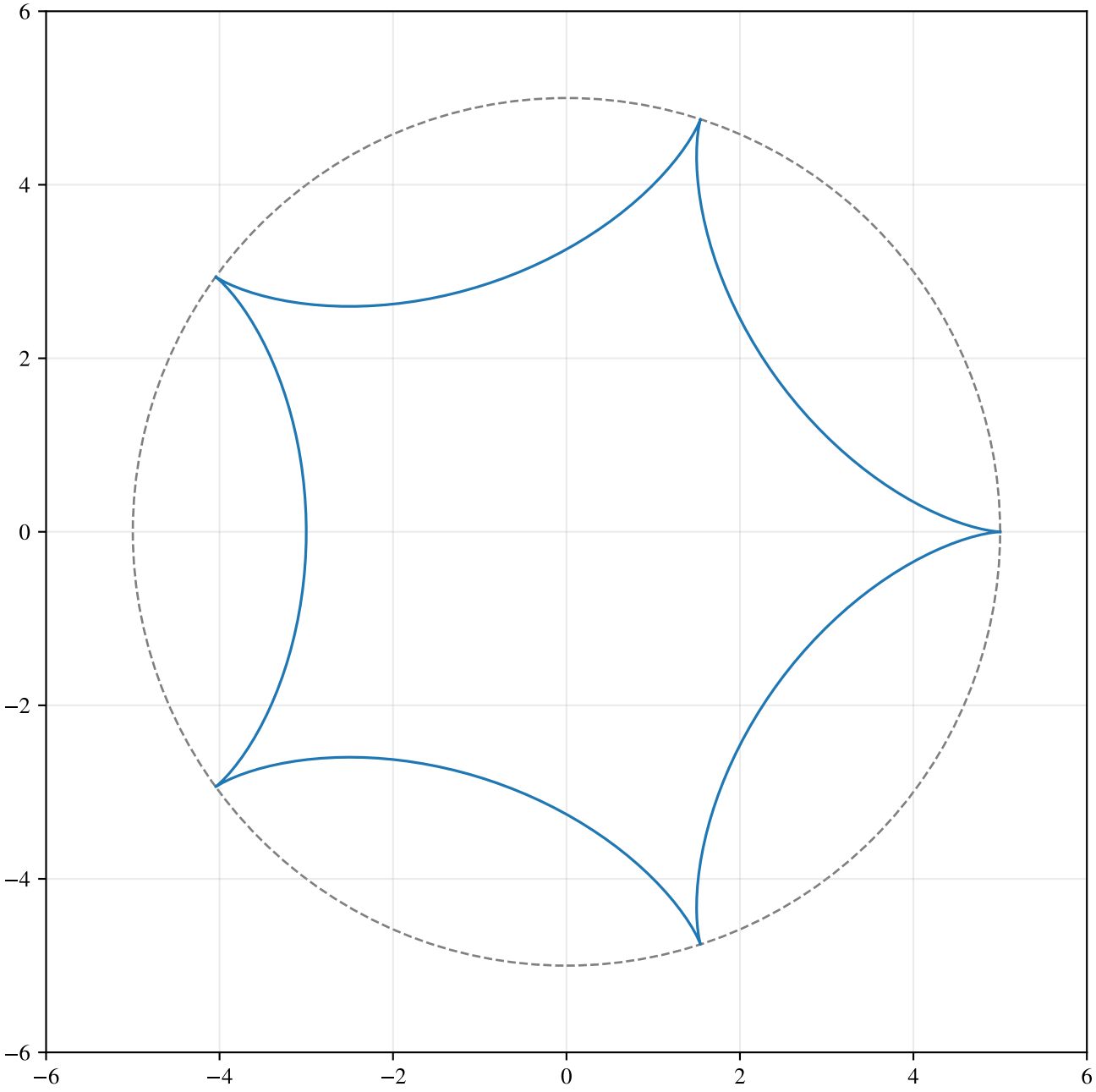
Entradas

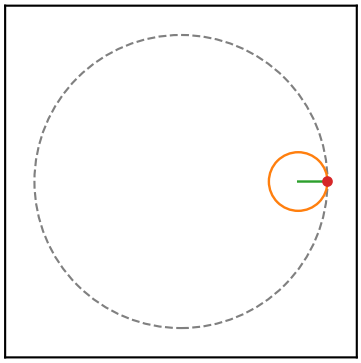
Parâmetro	Valor
R	5
r	1
RAZOES	3, 4, 5, 3.5
OMEGA	0.05236
INSTANTES	10, 30, 50
LAMBDAS	-1, 0.5, 1

1.1 — Hipocicloide

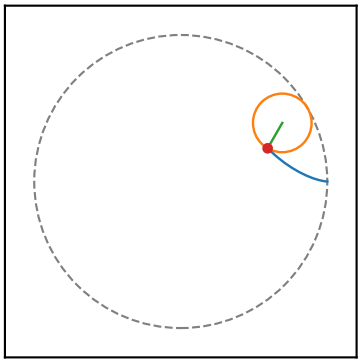


1.2 — Animação (quadros)

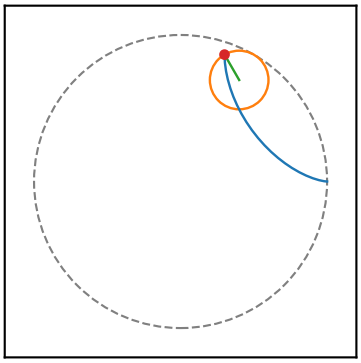
0.00



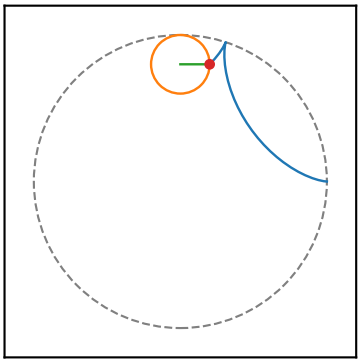
0.52



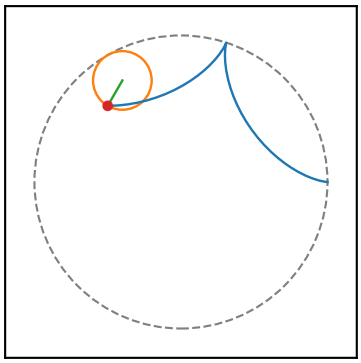
1.05



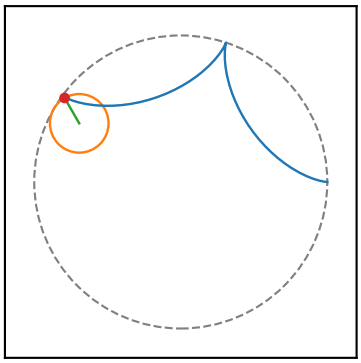
1.57



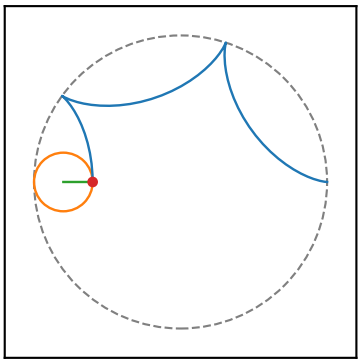
2.09



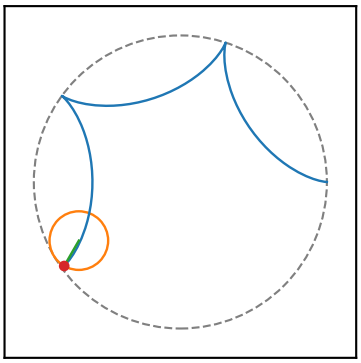
2.62



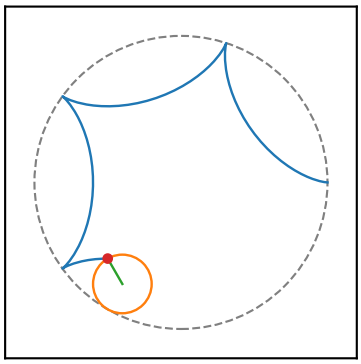
3.14



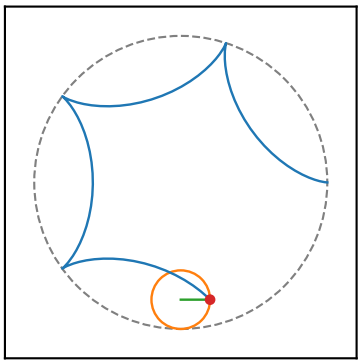
3.67



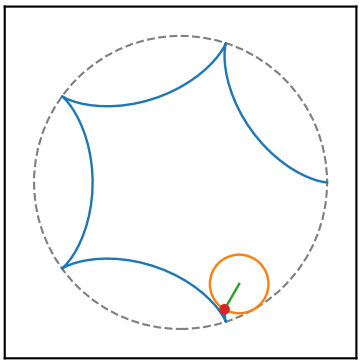
4.19



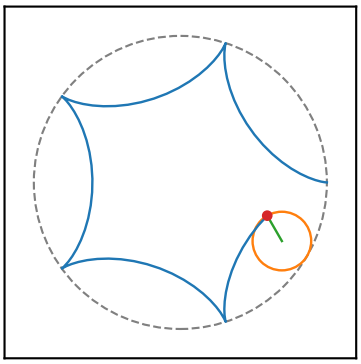
4.71



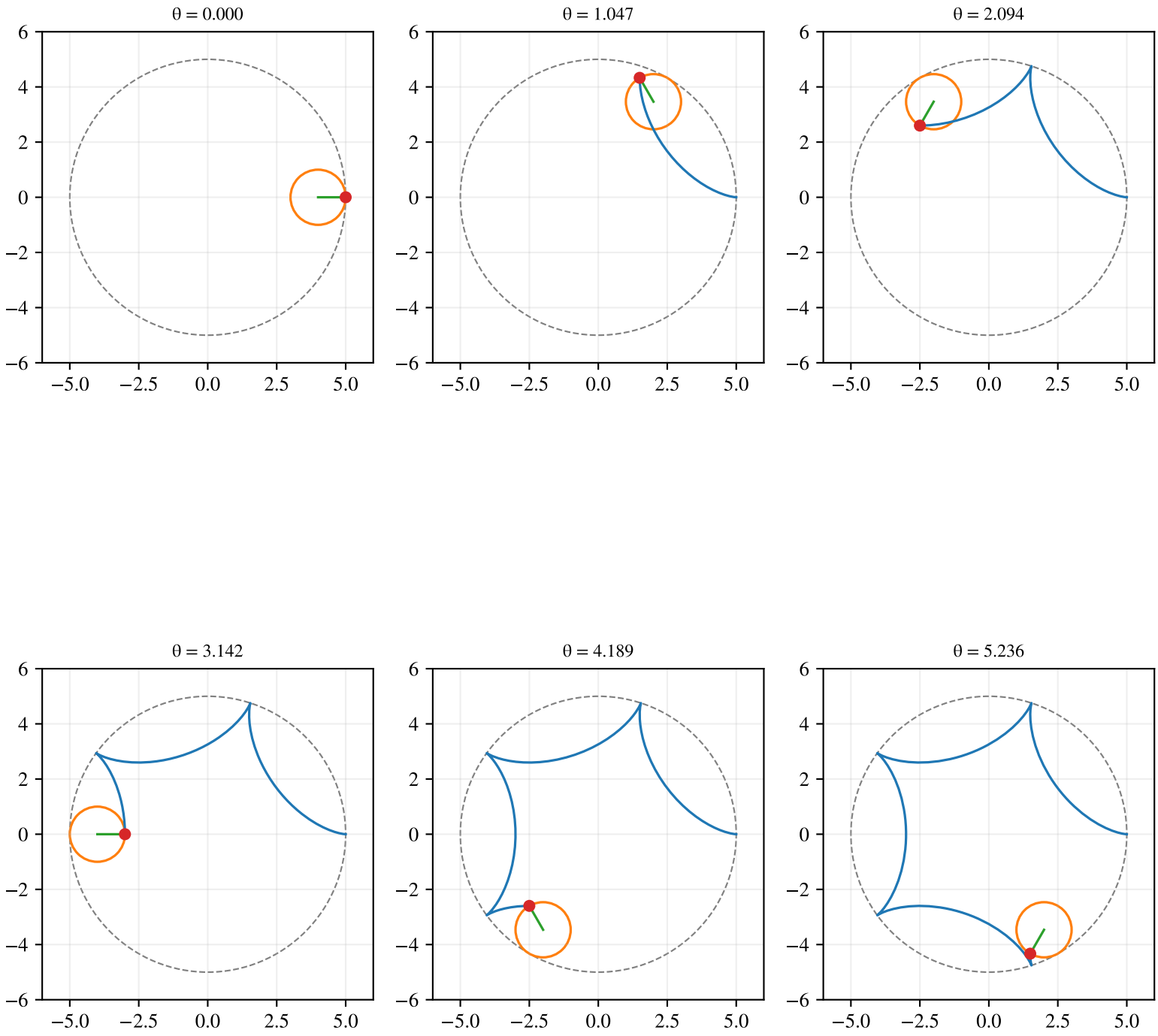
5.24



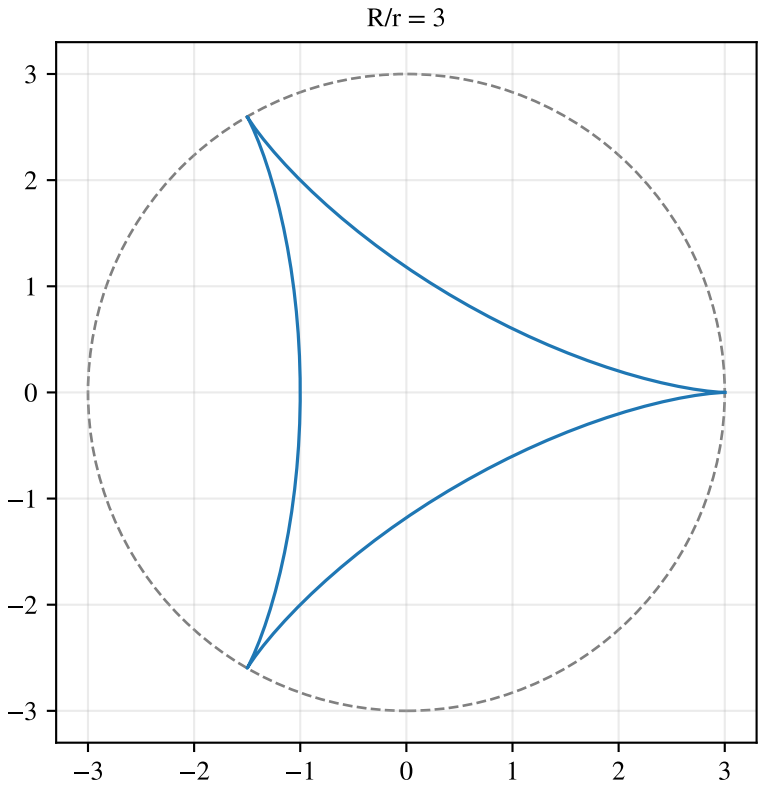
5.76

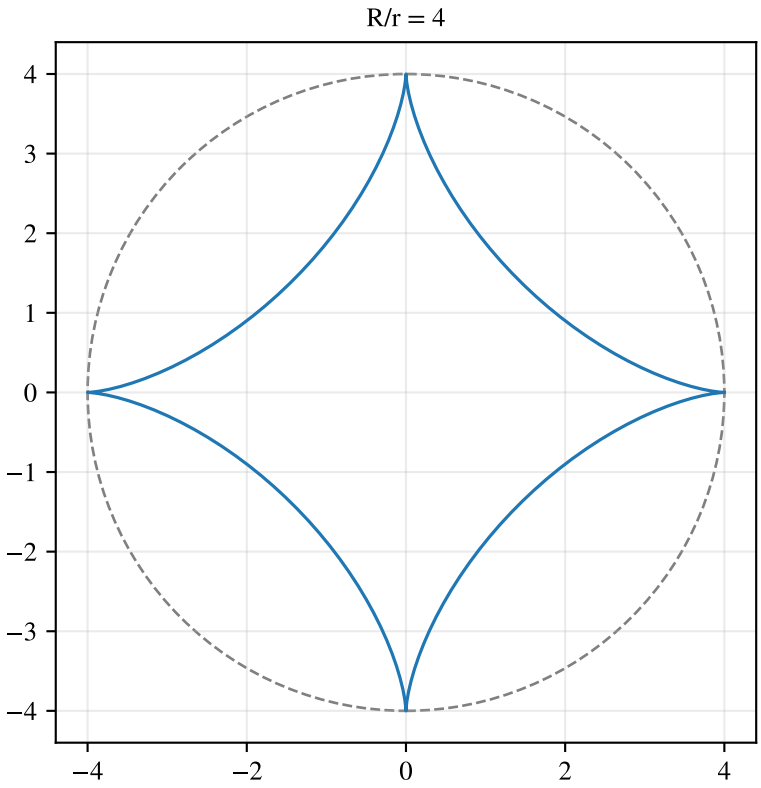


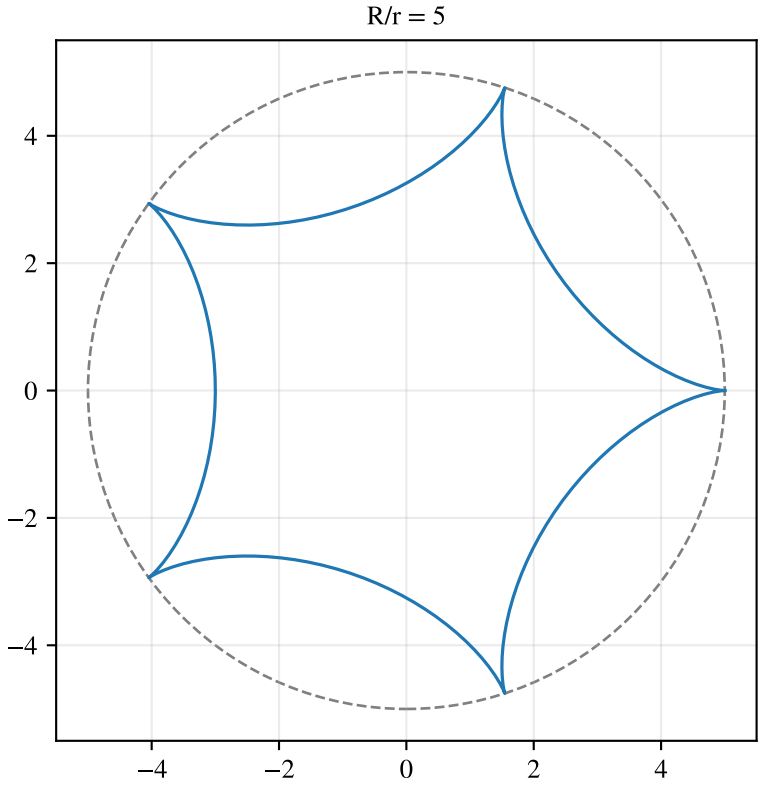
1.3 — Instantâneos

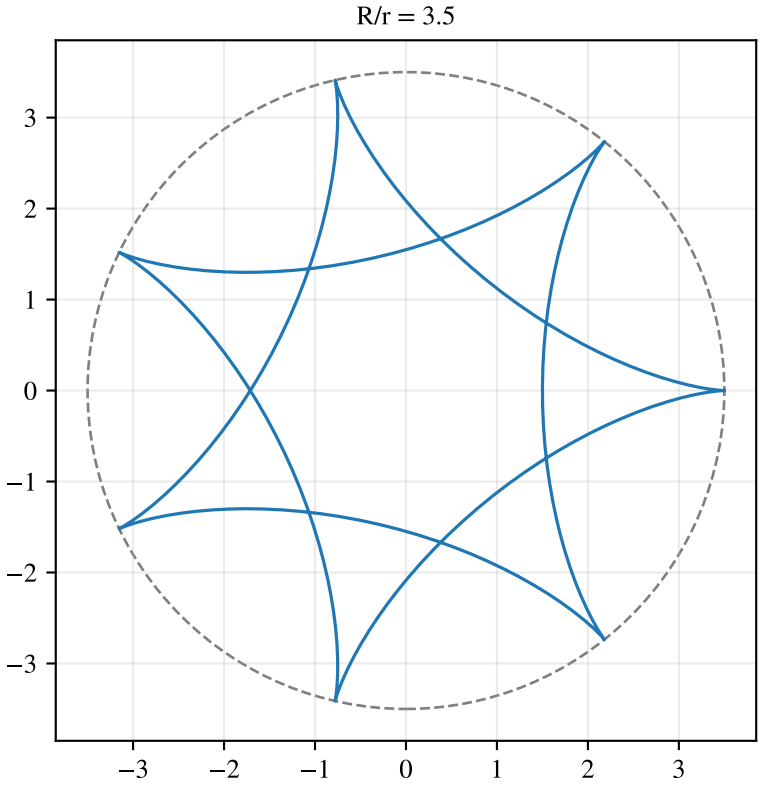


1.4 — Várias razões R/r

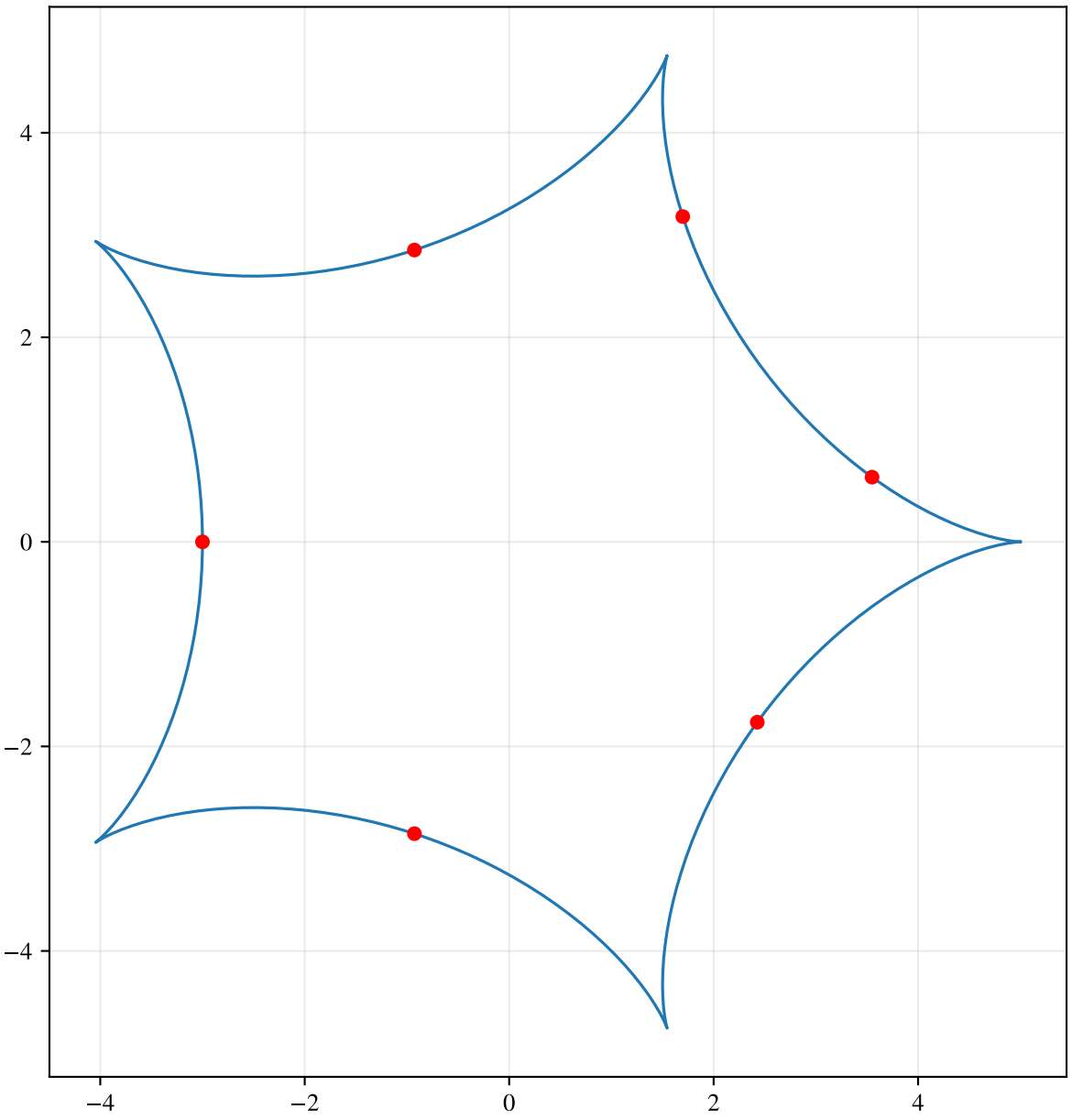






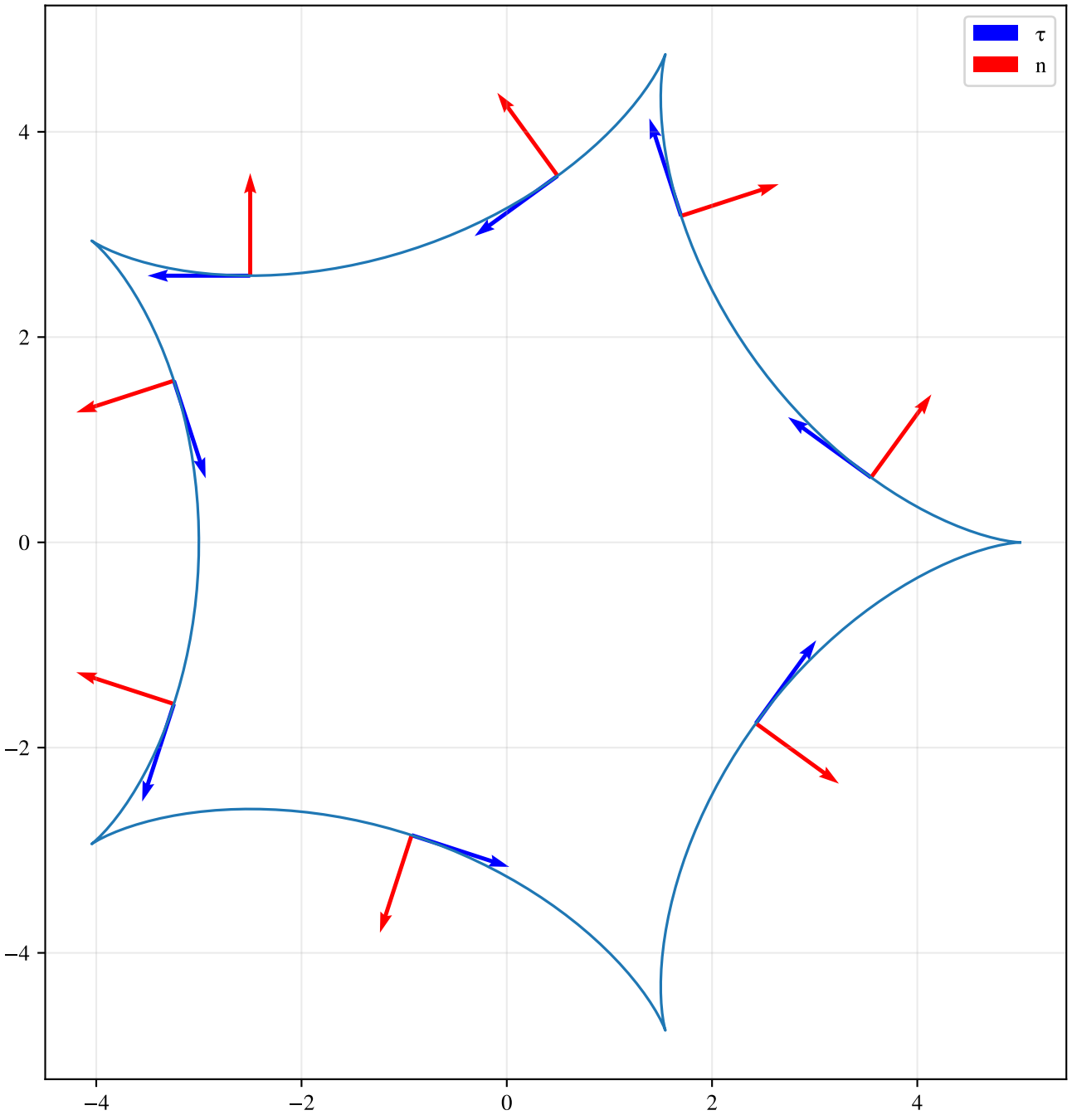


1.5 — Raio de curvatura

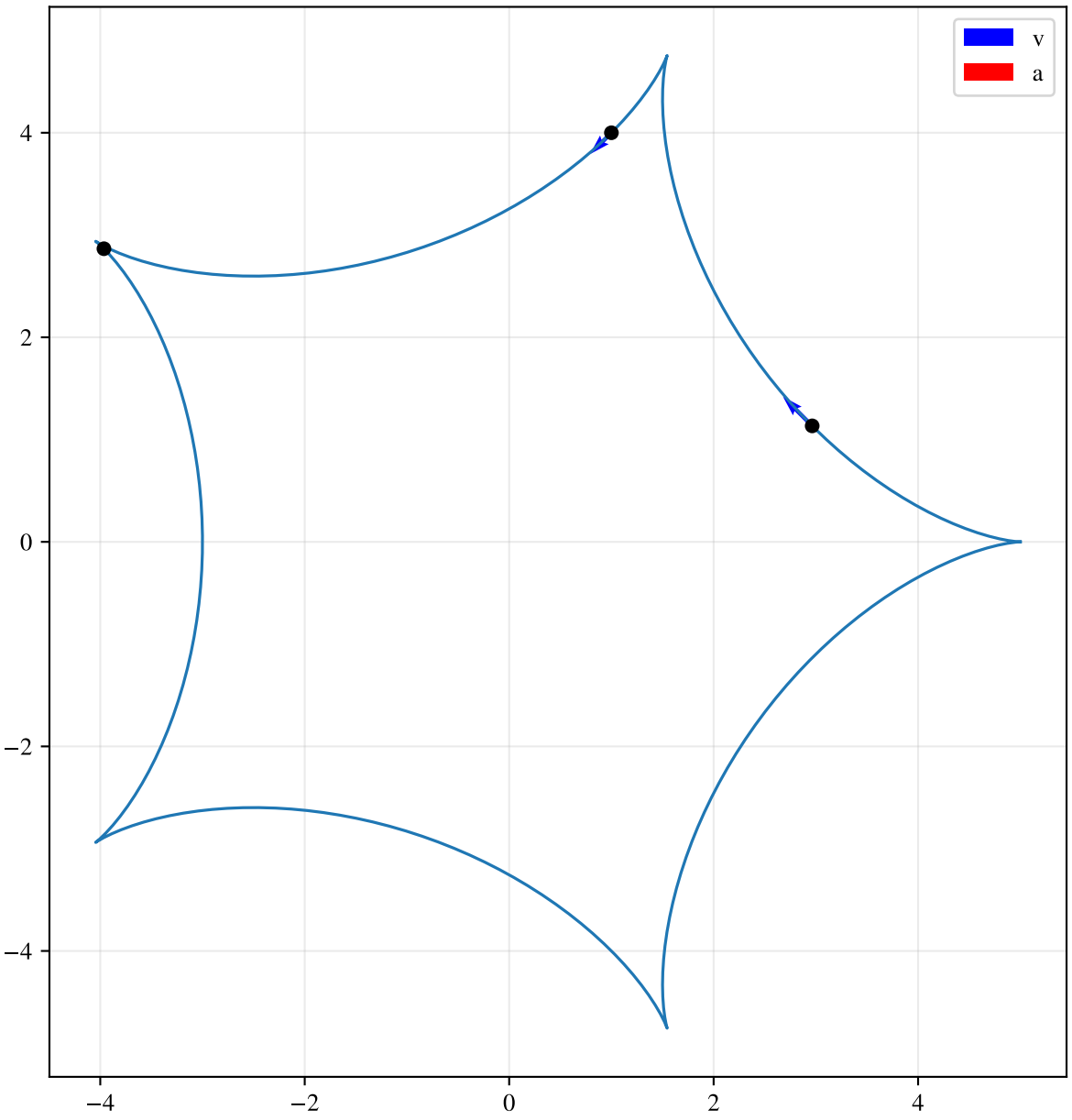


θ	x	y	ρ
0.4189	3.55	0.6324	4.619
0.8378	1.698	3.18	4.619
1.885	-0.9271	2.853	5.333
3.142	-3	9.797e-16	5.333
4.398	-0.9271	-2.853	5.333
5.655	2.427	-1.763	5.333

1.6 — Tangente e normal

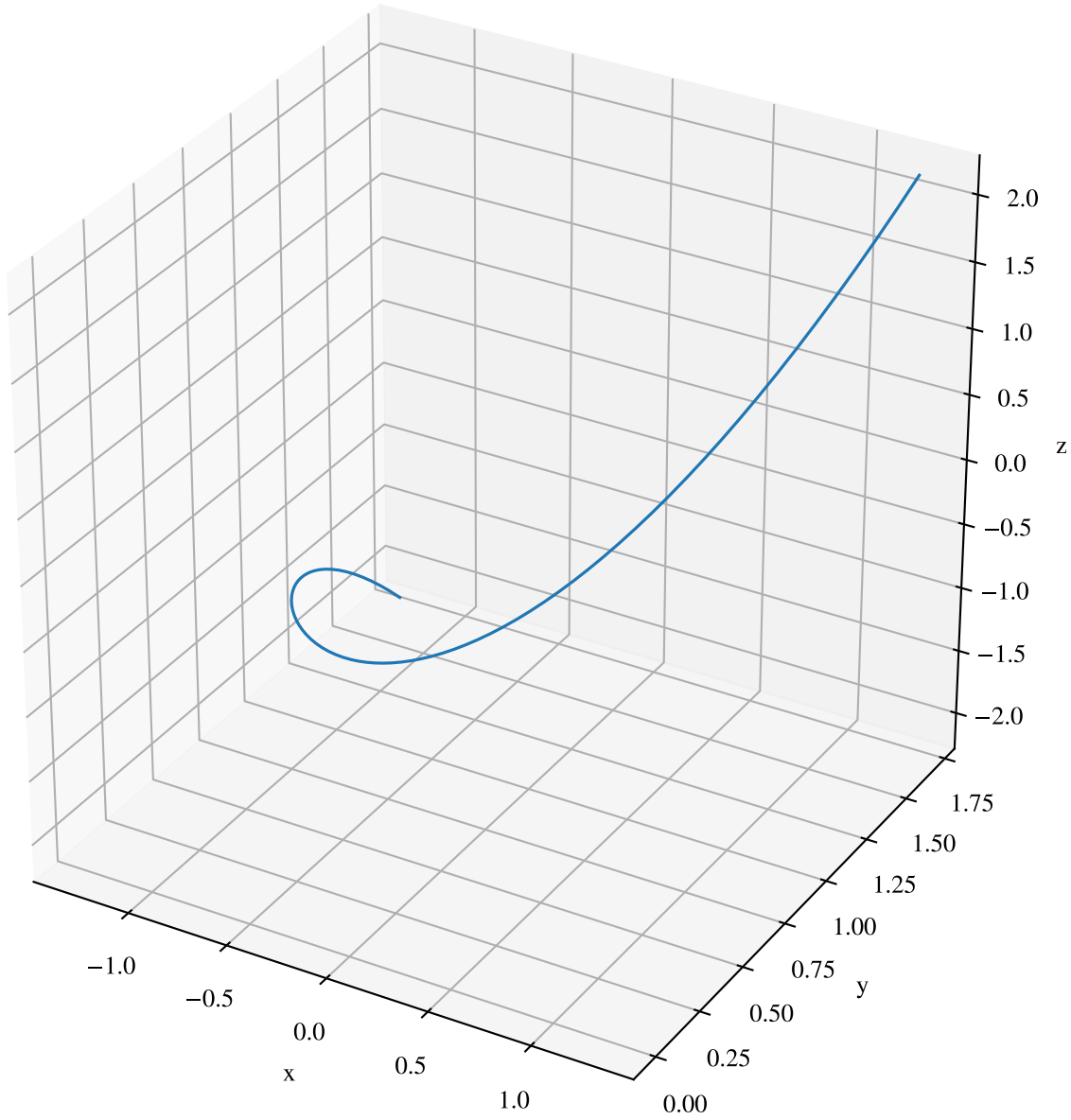


1.7 — Velocidade e aceleração



t	|v|	a_t	a_n	ρ
10	0.4046	0.01419	0.03178	5.152
30	0.2962	0.03877	0.02326	3.771
50	0.1084	0.05296	0.008515	1.38

2.1 — Curva espacial



λ	P=(x,y,z)	ρ
-1	(-1, 1, -1)	6.009
0.5	(0.5, 0.25, 0.125)	1.05
1	(1, 1, 1)	6.009

2.3 — Triedro de Frenet

